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SERIAL INTERFACE Using Python Software

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Presented here is a project for a simple transistor's output characteristic curve-tracer program through the serial interface and PIC microcontroller using Python programming language. Output characteristic curve of a transistor is automated using Python software and serial interface. Application software for collecting the measured value and displaying it on the monitor screen is developed using Python.

Engineers and academicians have been using high-level languages like Visual C++, Visual Basic and Delphi for connecting their gadgets/embedded systems with the computer for measurement and control purposes for a long time. Recently, many shifted their preference to Python as their computer language for measurement and analysis. The main advantage of using Python is the support of a large open source community.

Python is an interpreted programming language like BASIC, which saves considerable time during program development because compilation and linking are not required. Programs written in Python are typically much shorter as compared to ones written in C or C++. Python can be used for writing standalone programs and scripting applications. It is free, portable, powerful and remarkably easy to use.

To run a Python program, an interpreter is required. Python interpreter is a program that reads a Python program and then executes it. To write a Python program, it is advisable to use an editor (for typing a program). An editor is designed for writing Python programs. It provides highlighting and indentation that can help you write a program without mistake.

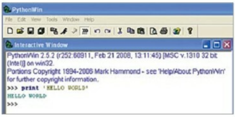

Fig. 1: Printing 'Hello World' on PythonWin window

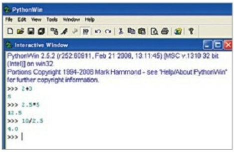

Fig. 2: Interactive Python Shell

The interpreter can be used to experiment with different features of the language. There are two modes in which interpreters can be used, namely, interactive and script. In interactive mode, the interpreter prints the result as the program is typed. The chevron, >>>, is the prompt the interpreter uses to indicate that it is ready.

Alternatively, you can store the code in a file and use the interpreter to execute the contents of the file, which is called a script. By convention, Python scripts end with .py extension. (Python installation package is available at *http://python.org*)

Writing the first Python program

After the installation of Python IDE and PyWin32 software, you can write a program and run it. The traditional first program is Hello World. This program simply prints 'Hello World' on the screen when it is run. This can be done with one statement in Python, as shown in Fig. 1.

PARTS LIST

Semiconductors:	
IC1	- PIC18F4620 microcontroller
IC2	- AD780 voltage regulator
IC3	- MAX5154 low-power, 12-bit DAC
IC4	- LM358 low-power dual op-amp
IC5	- MAX232 driver/receiver
LED1-LED2	- 5mm LED
D1-D3	- 1N4148 signal diode
T1	- BC107 npn transistor
T2	- SL100 npn power transistor
Resistors (all 1/4-watt, ±5% carbon):	
R1, R21	- 5.7-kilo-ohm
R2-R3	- 220-ohm
R4-R10	- 150-kilo-ohm
R11, R12	- 10-kilo-ohm
R13-R16	- 3.9-kilo-ohm
R17	- 47-kilo-ohm
R18	- 1-kilo-ohm
R19, R20	- 10-kilo-ohm
R22	- 1-ohm
Capacitors:	
C1-C4	- 0.1µF ceramic disk
Miscellaneous:	
CON1	- DE9 9-pin serial COM port male connector
CON2	- 2-pin connector for 12V battery
CON3	- 2-pin connector for 5V

Mathematical operations are performed in a single line without declaring the variables involved, as shown in Fig. 2.

Circuit and working

Circuit diagram of the serial interface and transistor curve tracer is shown in Fig. 3. The author's prototype is shown in Fig. 4. The circuit consists of a PIC18F4620 microcontroller (IC1), AD780 voltage regulator (IC2), MAX5154 12-bit digital-to-analogue converter, also called DAC (IC3), LM358 voltage amplifier (IC4), MAX232 dual driver/receiver (IC5), power transistor SL100/CL100 (T2), BC107 transistor-under-study (TUS) and other essential components needed for microcontroller operations.

Clock frequency required for microcontroller operation is derived